A Public Stagewise End-to-End Full-Key Recovery Attack on the `main.c` Implementation of SEPAR

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Abstract

We study the concrete SEPAR implementation in `SEPAR/main.c`, not an abstract idealized design. In the chosen-IV, chosen-plaintext, chosen-ciphertext, reset model, we give a public end-to-end full-key recovery attack on the 256-bit implementation key. The attack has a stochastic front-end and a deterministic exact core. The stochastic component is a chosen-IV search for suitable stage-8 and inward contexts. Once those IV sets are hit, every subsequent step is exact: an exact public stage-8 bootstrap for K_8; an exact stagewise inward search in which already-solved outer stages are peeled on fresh IVs by exact fixed-pair state recovery; and an exact stage-1 closure from the public stage-2 reduced table. Final acceptance is exact: a 256-bit key is accepted if and only if it reproduces the full 2^{16}-entry one-word oracle table on the successful IV. We define the exact attack primitives used by the implementation, prove conditional exactness, report complete recovery on difficult random keys, and give a full-program Monte Carlo over 100 pseudorandom 256-bit keys. In that experiment, the implemented attack succeeded on all 100 keys at inward budget 64 per stage, i.e. with total chosen-IV budget $512 + 32 + 6 \cdot 64 = 928$ IV contexts.

1 Introduction

The target is the concrete word-oriented implementation in `SEPAR/main.c`. The relevant attack surface is the resettable 16-bit one-word oracle induced by that code. For every fixed IV, the implementation exposes a public permutation

$$F_{\mathrm{IV}} : \mathbb{Z}_{2^{16}} \to \mathbb{Z}_{2^{16}}$$

on one-word inputs. Because the block size is only 16 bits, the entire codebook is observable with 2^{16} chosen encryptions and can then be manipulated exactly.

The attack model is:

- chosen IV;
- chosen plaintext and chosen ciphertext;
- exact state reset between oracle calls.

Under that model, the implementation gives exact public access to

$$F_{\mathrm{IV}}(x) = \mathrm{Enc}_{K,\mathrm{IV}}(x), \qquad F_{\mathrm{IV}}^{-1}(y) = \mathrm{Dec}_{K,\mathrm{IV}}(y),$$

and the implementation used in the attack constructs the inverse locally by inverting the public encryption table.

Main claim. The public attack implemented in

`separ_public_stagewise_attack.c`

recovers the full 256-bit key by:

1. a chosen-IV beam search and exact hybrid public bootstrap for K_8;
2. an exact stagewise inward search for $K_7, K_6, \ldots, K_2$ in which solved outer stages are peeled on fresh IVs by exact fixed-pair state recovery;
3. an exact stage-1 closure from the public stage-2 reduced table;
4. exact full-codebook verification of the recovered 256-bit key.

What is proved below is *conditional exactness*: once the sampled IV sets contain the true exact candidates required by the attack, the remainder of the recovery is deterministic and exact.

2 Exact implementation model

2.1 Stage-local permutation

Let the eight 32-bit stage keys be

$$K_i = (k_{i,0}, k_{i,1}) \in \mathbb{Z}_{2^{16}} \times \mathbb{Z}_{2^{16}}, \qquad 1 \le i \le 8.$$

Let $\Sigma : \mathbb{Z}_{2^{16}} \to \mathbb{Z}_{2^{16}}$ be the nibblewise S-box layer induced by the fixed tables S_1, S_2, S_3, S_4 in `main.c`, and Σ^{-1} the inverse layer induced by `IS1`, ..., `IS4`.

Let $L : \mathbb{Z}_{2^{16}} \to \mathbb{Z}_{2^{16}}$ be the exact linear layer implemented by `sep_rotl16`. If

$$x = (a, b, c, d) \in (\mathbb{Z}_{16})^4$$

in nibble notation, define

$$M(a, b, c, d) = (a \oplus c,\ b \oplus d,\ c \oplus b,\ d \oplus a),$$

identify nibble tuples with 16-bit words, and set

$$L(x) = M(x) \oplus \mathrm{rotl}_{16}(M(x), 12) \oplus \mathrm{rotl}_{16}(M(x), 8).$$

Its inverse is exactly `sep_inrotl16`.

For each stage i, the code derives two further 16-bit subkeys $\kappa_{i,2}, \kappa_{i,3}$ from $(k_{i,0}, k_{i,1})$:

$$\begin{aligned} u_i &= \mathrm{rotl}_{16}(k_{i,0}, 6), & v_i &= \mathrm{rotl}_{16}(k_{i,1}, 10), \\ \kappa_{i,2} &= \big(u_i \vee (S_1((u_i \gg 6) \bmod 16) \ll 6)\big) \oplus (i+2), \\ \kappa_{i,3} &= \big(v_i \vee (S_1((v_i \gg 6) \bmod 16) \ll 6)\big) \oplus (i+3). \end{aligned}$$

With $A_t(x) = x \oplus t$, the stage-local encryption permutation implemented by `enc_block` is exactly

$$E_i = A_{\kappa_{i,2} \oplus \kappa_{i,3}} \circ \Sigma \circ A_{k_{i,0} \oplus k_{i,1}} \circ L \circ \Sigma \circ A_{\kappa_{i,3}} \circ L \circ \Sigma \circ A_{\kappa_{i,2}} \circ L \circ \Sigma \circ A_{k_{i,1}} \circ L \circ \Sigma \circ A_{k_{i,0}}, \quad (1)$$

and $D_i = E_i^{-1}$ is exactly `dec_block`.

2.2 Translated stage action and one-word encryption

For a state word $s_i \in \mathbb{Z}_{2^{16}}$, define the translated stage permutation

$$B_i^{(s_i)}(x) = E_i(x + s_i), \qquad \big(B_i^{(s_i)}\big)^{-1}(y) = D_i(y) - s_i,$$

where all additions and subtractions are in $\mathbb{Z}_{2^{16}}$.

For a reset IV with state $(s_1, \ldots, s_8)$, the public one-word encryption permutation is

$$F_{\mathrm{IV}} = B_8^{(s_8)} \circ B_7^{(s_7)} \circ \cdots \circ B_1^{(s_1)}.$$

2.3 State update law

Write the eight state words as $(s_1, \ldots, s_8)$ and the intermediate stage outputs as

$$v_{12}, v_{23}, v_{34}, v_{45}, v_{56}, v_{67}, v_{78}, c.$$

From `Separ_Encryption` in `main.c`, the exact state update after one word is

$$s_2' = s_2 + v_{12} + v_{56} + s_6, \tag{2}$$
$$s_3' = s_3 + v_{23} + v_{34} + s_4 + s_1, \tag{3}$$
$$s_4' = s_4 + v_{12} + v_{45} + s_8, \tag{4}$$
$$s_5' = s_5 + v_{23} + L', \tag{5}$$
$$s_6' = s_6 + v_{12} + v_{45} + s_7, \tag{6}$$
$$s_7' = s_7 + v_{23} + v_{67}, \tag{7}$$
$$s_8' = s_8 + v_{45}, \tag{8}$$
$$s_1' = s_1 + s_5 + v_{34} + 2v_{23} + v_{78}. \tag{9}$$

The attack below uses only reset one-word tables, so these equations matter only because they determine which state word appears as the output translation at each stage.

3 Exact public tables used by the attack

3.1 Recursive current tables

For $m \in \{1, \ldots, 7\}$ define the inner cascade

$$R_m := B_m^{(s_m)} \circ B_{m-1}^{(s_{m-1})} \circ \cdots \circ B_1^{(s_1)}.$$

Definition 1 (Current table at level m)**.** *Assume $K_{m+1}, \ldots, K_8$ and $s_{m+2}, \ldots, s_8$ are known for a fixed IV. The current table at level m is the public table*

$$C_m : \mathbb{Z}_{2^{16}} \to \mathbb{Z}_{2^{16}}$$

computed recursively by

$$C_7 = D_8 \circ F_{\mathrm{IV}}, \qquad C_m = D_{m+1}\big(C_{m+1} - s_{m+2}\big) \quad (1 \le m \le 6).$$

Lemma 1 (Exact translation form)**.** *For every $m \in \{1, \ldots, 7\}$,*

$$C_m(x) = R_m(x) + s_{m+1} \qquad \textit{for all } x \in \mathbb{Z}_{2^{16}}.$$

Proof. For $m = 7$,

$$F_{\mathrm{IV}}(x) = E_8(R_7(x) + s_8),$$

hence

$$C_7(x) = D_8(F_{\mathrm{IV}}(x)) = R_7(x) + s_8.$$

Assume the statement holds at level $m + 1$. Then

$$C_{m+1}(x) = R_{m+1}(x) + s_{m+2} = E_{m+1}(R_m(x) + s_{m+1}) + s_{m+2},$$

so

$$C_m(x) = D_{m+1}(C_{m+1}(x) - s_{m+2}) = R_m(x) + s_{m+1}.$$

□

Thus the exact public object attacked at level m is the inner cascade R_m observed through a single output translation by s_{m+1}.

3.2 Exact low-byte score

Let

$$H(y) = \left\lfloor \frac{y}{2^8} \right\rfloor \in \mathbb{Z}_{256}$$

be the output high byte. For a public table $T : \mathbb{Z}_{2^{16}} \to \mathbb{Z}_{2^{16}}$, a candidate low byte $b \in \mathbb{Z}_{256}$, and a row index $r \in \mathbb{Z}_{256}$, define

$$\sigma_{T,b}(r) := \left|\left\{H\big(T(256r + \lambda) - b\big) : \lambda \in \mathbb{Z}_{256}\right\}\right|,$$

and the exact total score

$$\Sigma(T; b) := \sum_{r=0}^{255} \sigma_{T,b}(r).$$

This is exactly the quantity computed by `exact_low_byte_scan_u16()`.

3.3 Projected quotient cycles, representatives, and verifier

Define the 4-bit projection

$$\mathrm{proj}(y) := H(y) \bmod 16 \in \mathbb{Z}_{16}.$$

For a stage key pair K_i, define the stage-i quotient map

$$q_{i,K_i}(h) = \mathrm{proj}(E_i(256h)), \qquad h \in \mathbb{Z}_{16},$$

and the full high-byte quotient map

$$Q_{i,K_i}(h) = H(E_i(256h)), \qquad h \in \mathbb{Z}_{256}.$$

The implementation uses the exact auxiliary sets

$$\Delta_0 = \{1, 2, 4, 8, 15, 16\},$$

$$\Omega = \{0, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 200, 233, 255\},$$

$$\mathcal{R} = \{0x00, 0x55, 0xAA, 0xFF\}.$$

The corrected implementation uses the same universal invisible-bit set for every stage $i \in \{2, \ldots, 8\}$:

$$P_i = \{6, 7, 12, 13, 14, 15, 18, 19, 20, 21, 22, 23\}.$$

Hence

$$|P_i| = 12, \qquad |V_i| = 20,$$

and every representative table enumerates exactly

$$2^{20} = 1{,}048{,}576$$

visible masks. Stages $2, \ldots, 7$ canonicalize 16-cycles by rotation to first entry 0; stage 8 uses raw cycles.

For any public table T, define the exact projected edge-count matrix

$$M_T(a,b) := \left|\{(x,\delta) \in \mathbb{Z}_{2^{16}} \times \Delta_0 : \mathrm{proj}(T(x)) = a,\ \mathrm{proj}(T(x+\delta)) = b\}\right|, \qquad a,b \in \mathbb{Z}_{16}.$$

If $c = (c_0, \ldots, c_{15})$ is a 16-cycle, define the exact cycle score

$$S_T(c) = \sum_{j=0}^{15} M_T(c_j, c_{j+1 \bmod 16}).$$

To lift a projected cycle to a 256-point high-byte order, the implementation builds the exact 8-bit transition matrix

$$N_T(u,v) := \left|\{(x,\delta) \in \mathbb{Z}_{2^{16}} \times \Delta_0 : H(T(x)) = u,\ H(T(x+\delta)) = v\}\right|, \qquad u,v \in \mathbb{Z}_{256},$$

solves exact 16×16 assignments between adjacent cycle layers by the Hungarian algorithm, and accepts only wrap permutations forming one 16-cycle.

Given a recovered cycle/order pair, the implementation enumerates every compatible representative-group base key and expands all invisible masks. It then checks exact equality against the reconstructed order targets first on Ω and then on all 256 values. Finally it exact-scores the surviving candidates by

$$V_i(K_i; T) := \sum_{r \in \mathcal{R}} \min_{b \in \mathbb{Z}_{256}} \left|\left\{H\big(D_i(T(256r+\lambda)) - b\big) : \lambda \in \mathbb{Z}_{256}\right\}\right|.$$

This is the verifier used both in the outer bootstrap and in the stagewise inward search.

4 The outer public bootstrap for K_8

4.1 Weak-IV beam search

For an IV, define the exact row score

$$\rho(\mathrm{IV}) := \left|\{H(F_{\mathrm{IV}}(\lambda)) : \lambda \in \mathbb{Z}_{256}\}\right|.$$

The implementation keeps the B IVs with smallest $\rho(\mathrm{IV})$. This is a search-ordering device only; it is not part of the acceptance criterion.

4.2 Hybrid exact stage-8 recovery

For each IV in the beam, the implementation:

1. computes the exact stage-8 quotient/bootstrap candidate set by projected-cycle maximization and exact verifier scoring;
2. closes that set under the independent MSB toggles
$$(a,b) \mapsto (a \oplus \epsilon_0 2^{15},\ b \oplus \epsilon_1 2^{15});$$
3. exact-scores the resulting small pool by the additive differential score
$$A_{\mathrm{IV}}(K_8) := \sum_{\delta \in \Delta_0} \max_{\Delta \in \mathbb{Z}_{2^{16}}} \left|\{x : D_8(F_{\mathrm{IV}}(x+\delta); K_8) - D_8(F_{\mathrm{IV}}(x); K_8) = \Delta\}\right|;$$
4. aggregates those exact scores across the beam and outputs the best K_8.

5 Exact stagewise inward search from K_7 to K_2

5.1 Exact fixed-pair peeling on fresh IVs

The inward phase is not a one-IV cascade. Once the outer bootstrap has fixed K_8, the program samples a fresh IV set independently at each target stage. On each such IV it peels every already-solved outer stage by exact state recovery for the fixed known stage key, and only then attacks the next unknown stage.

Fix a target stage $t \in \{7, 6, 5, 4, 3, 2\}$ and suppose a branch has already fixed

$$K_{t+1}, K_{t+2}, \ldots, K_8.$$

Let η be a fresh IV. Define

$$C_7^{(\eta)} := D_8 \circ F_\eta.$$

If K_8 is correct, then by Lemma 1,

$$C_7^{(\eta)}(x) = R_7^{(\eta)}(x) + s_8^{(\eta)}.$$

For a fixed stage $m \in \{7, \ldots, t+1\}$ and a public table T, define the minimizing low-byte set

$$L_m(T) := \arg\min_{b \in \mathbb{Z}_{256}} \Sigma(T; b).$$

For a fixed known pair K_m, define the exact fixed-pair state solver

$$\mathcal{S}_m(T; K_m) := \arg \min_{\substack{s=b+256h \\ b \in L_m(T),\, h \in \mathbb{Z}_{256}}} V_m(K_m; T - s), \tag{10}$$

with lexicographic tie-break on s. This is exactly `recover_known_pair_state_fast()`.

Starting from $C_7^{(\eta)}$, the program peels already-solved outer stages by

$$\widehat{s}_{m+1}^{(\eta)} := \mathcal{S}_m(C_m^{(\eta)}; K_m), \tag{11}$$

$$C_{m-1}^{(\eta)} := D_m\left(C_m^{(\eta)} - \widehat{s}_{m+1}^{(\eta)}; K_m\right), \qquad m = 7, 6, \ldots, t+1. \tag{12}$$

If every known outer pair on the current branch is correct and every fixed-pair state recovery returns the true state word, then

$$C_t^{(\eta)}(x) = R_t^{(\eta)}(x) + s_{t+1}^{(\eta)}.$$

5.2 Exact target-stage candidate sets

For a fixed target stage t and fresh IV η, let

$$\mathcal{A}_t(\eta; K_{t+1}, \ldots, K_8)$$

denote the set of all minimum-verifier rows produced by applying the exact stage-t scan of Sections 3.2–3.4 to the prepared table $C_t^{(\eta)}$. Every element of $\mathcal{A}_t$ is a pair

$$(K_t, s_{t+1}) \in (\mathbb{Z}_{2^{16}})^2.$$

Write

$$\Pi_{\text{pair}} \mathcal{A}_t(\eta; K_{t+1}, \ldots, K_8)$$

for the set of stage-key pairs appearing in those rows.

Let

$$\Omega_t(N) = \{\eta_{t,1}, \ldots, \eta_{t,N}\}$$

be the first N fresh IVs in the deterministic stage-t IV stream. Define the exact stagewise candidate-pair set

$$\mathcal{P}_t^{(N)}(K_{t+1}, \ldots, K_8) := \bigcup_{\eta \in \Omega_t(N)} \Pi_{\text{pair}} \mathcal{A}_t(\eta; K_{t+1}, \ldots, K_8). \tag{13}$$

The implementation also records the exact hit multiplicity

$$h_t^{(N)}(K_t) := \left|\{\eta \in \Omega_t(N) : K_t \in \Pi_{\text{pair}} \mathcal{A}_t(\eta; K_{t+1}, \ldots, K_8)\}\right|, \tag{14}$$

sorts candidates by decreasing $h_t^{(N)}$, and recursively explores all distinct pairs in that exact order. This ordering affects runtime only: no candidate in (13) is discarded.

5.3 Exact recursive search tree

Define recursively the exact search procedures

$$\mathfrak{S}_t^{(N)}(K_{t+1}, \ldots, K_8), \qquad t = 7, 6, \ldots, 2,$$

as follows.

For $t \in \{7, 6, 5, 4, 3\}$:

1. build $\mathcal{P}_t^{(N)}(K_{t+1}, \ldots, K_8)$ by (13);
2. enumerate all $K_t \in \mathcal{P}_t^{(N)}$ in decreasing order of (14);
3. recurse to
$$\mathfrak{S}_{t-1}^{(N)}(K_t, K_{t+1}, \ldots, K_8).$$

For $t = 2$, there is no further pair recursion. Instead, for every $\eta \in \Omega_2(N)$ the program:

1. prepares the exact public stage-2 current table $C_2^{(\eta)}$ by fixed-pair peeling of the already-solved suffix;
2. computes the exact minimum-verifier row set $\mathcal{A}_2(\eta; K_3, \ldots, K_8)$;
3. for every row $(K_2, s_3) \in \mathcal{A}_2(\eta; K_3, \ldots, K_8)$, invokes the exact stage-1 closure of Section 6;
4. accepts only if the recovered 256-bit key reproduces the entire 2^{16}-entry one-word oracle table on the same IV.

Why stage 1 is separate. The projected-family scan exists only for stages $2, \ldots, 8$. Stage 1 has no visible successor stage and therefore no stage-1 quotient-cycle analogue. It is closed exactly from the stage-2 reduced table.

Proposition 1 (Monotonicity in the inward budget)**.** *Fix a key, the outerly recovered K_8, and the deterministic per-stage IV streams. If the stagewise attack succeeds with inward budget N, then it also succeeds with every larger budget $N' \geq N$.*

Proof. For every stage t, the sampled IV set $\Omega_t(N)$ is a prefix of $\Omega_t(N')$. Hence

$$\mathcal{P}_t^{(N)}(K_{t+1}, \ldots, K_8) \subseteq \mathcal{P}_t^{(N')}(K_{t+1}, \ldots, K_8)$$

for every fixed suffix. Thus every branch explored at budget N is also explored at budget N'. At stage 2 the tested IV set is likewise prefix-monotone, so any successful stage-2 IV at budget N is still tested at budget N'. Therefore success is monotone. □

6 Exact stage-1 closure from the public stage-2 table

6.1 Reduction to a single translated stage

Assume the stagewise recursion has recovered the true (K_2, s_3) on some successful stage-2 IV. Then

$$C_2(x) = E_2(E_1(x + s_1) + s_2) + s_3.$$

Define

$$W_2(x) = D_2(C_2(x) - s_3; K_2).$$

Then exactly

$$W_2(x) = E_1(x + s_1) + s_2. \tag{15}$$

6.2 Low-byte recovery for s_2

The implementation first applies the same exact low-byte scan to W_2. For each candidate

$$s_2^\star \in \mathbb{Z}_{2^{16}},$$

it forms

$$C^\star(x) = W_2(x) - s_2^\star.$$

Any true stage-1 solution must satisfy

$$C^\star(x) = E_1(x + s_1; K_1).$$

6.3 Step 1: exact row-signature inversion

For $u \in \mathbb{Z}_{256}$ define the 16-point row signature

$$R_u^{C^\star}(h) = \mathrm{proj}(C^\star(u + 256h)), \qquad h \in \mathbb{Z}_{16}.$$

Write

$$s_1 = s_{1,\mathrm{lo}} + 256\, s_{1,\mathrm{mid}} + 4096\, s_{1,\mathrm{hi}},$$

with $s_{1,\mathrm{lo}} \in \mathbb{Z}_{256}$ and $s_{1,\mathrm{mid}}, s_{1,\mathrm{hi}} \in \mathbb{Z}_{16}$.

For a candidate $s_{1,\mathrm{lo}}$, the routine `stage1_step1_candidates_u16()` checks the exact carry partition:

> all rows with $u + s_{1,\mathrm{lo}} < 256$ must share one common signature σ, and all rows with $u + s_{1,\mathrm{lo}} \geq 256$ must share the one-step rotation $\mathrm{rot}_1(\sigma)$.

For each surviving $s_{1,\mathrm{lo}}$ and each $s_{1,\mathrm{mid}} \in \mathbb{Z}_{16}$, the implementation rotates σ by $16 - s_{1,\mathrm{mid}}$ and inverts it through the exact precomputed family

$$\nu_{a,b,c,d} : \mathbb{Z}_{16} \to \mathbb{Z}_{16},$$

where

$$\nu_{a,b,c,d}(x) = S_2\big(t_4(x) \oplus a \oplus b\big) \oplus c \oplus d.$$

with

$$\begin{aligned} t_1(x) &= S_2(x \oplus a), \\ t_2(x) &= S_2(t_1(x) \oplus b), \\ t_3(x) &= S_2(t_2(x) \oplus c), \\ t_4(x) &= S_2(t_3(x) \oplus d). \end{aligned}$$

This yields exact candidates for

$$s_{1,\mathrm{lo}}, \quad s_{1,\mathrm{mid}}, \quad a, \quad b, \quad c, \quad d.$$

6.4 Step 2: exact lift to full K_1 candidates

The routine next uses exact inverse-image tables for the derived-key high nibbles of $\kappa_{1,2}$ and $\kappa_{1,3}$. For each surviving $(s_{1,\mathrm{lo}}, s_{1,\mathrm{mid}}, a, b, c, d)$, each

$$\ell_6 \in I_2(c), \qquad \beta \in I_3(d),$$

and each 12-bit free mask $f \in \mathbb{Z}_{2^{12}}$, it constructs

$$k_{1,0} = \ell_6 \ \vee \ \big((f \bmod 4) \ll 6\big) \ \vee \ (a \ll 8) \ \vee \ \left(\left(\left\lfloor \frac{f}{4} \right\rfloor \bmod 16\right) \ll 12\right), \tag{16}$$

$$k_{1,1} = (\beta \bmod 4) \ \vee \ \left(\left(\left\lfloor \frac{f}{64} \right\rfloor \bmod 64\right) \ll 2\right) \ \vee \ (b \ll 8) \ \vee \ \left(\left(\left\lfloor \frac{\beta}{4} \right\rfloor \bmod 16\right) \ll 12\right). \tag{17}$$

The resulting pair is exact-tested on the 12-bit slice

$$X_{12} = \{0, \ldots, 4095\}$$

by comparing nibble (7..4) values:

$$\left\lfloor \frac{E_1((x + s_{1,\mathrm{lo}} + 256 s_{1,\mathrm{mid}}) \bmod 2^{12}; K_1)}{2^4} \right\rfloor \bmod 16 = \left\lfloor \frac{C^\star(x)}{2^4} \right\rfloor \bmod 16 \qquad \text{for all } x \in X_{12}.$$

6.5 Step 3: exact recovery of the top nibble of s_1

For each surviving pair K_1, the implementation scans all 16 possibilities for $s_{1,\mathrm{hi}}$ and performs full exact verification:

$$E_1(x + s_1; K_1) = C^\star(x) \qquad \text{for all } x \in \mathbb{Z}_{2^{16}}.$$

If exactly one triple (K_1, s_1, s_2) survives, the stage-1 closure succeeds.

Theorem 1 (Exactness of the stage-1 closure)**.** *Let W_2 be the exact public table of* (15)*. If there is a unique triple (K_1, s_1, s_2) satisfying*

$$W_2(x) = E_1(x + s_1; K_1) + s_2 \qquad \textit{for all } x \in \mathbb{Z}_{2^{16}},$$

then `stage1_finish_from_stage2_current()` *returns exactly that triple.*

Proof. The routine is exhaustive over all candidates for the low and high bytes of s_2, all exact row-signature solutions for $(s_{1,\mathrm{lo}}, s_{1,\mathrm{mid}}, a, b, c, d)$, all inverse-image candidates from $I_2(c)$ and $I_3(d)$, all 2^{12} free masks in (16)–(17), and all 16 possibilities for $s_{1,\mathrm{hi}}$. Every surviving candidate is tested by exact equality either on the 12-bit slice or on the full 2^{16}-entry table, and the routine accepts only if the number of full solutions is exactly one. Hence a unique solution is returned exactly. □

7 Public full-key acceptance and conditional correctness

Definition 2 (Exact public acceptance test)**.** *Given recovered pairs*

$$\widehat{K} = (\widehat{K}_1, \ldots, \widehat{K}_8),$$

rebuild the candidate one-word table

$$\widehat{F}_{\mathrm{IV}}$$

under the same IV and accept if and only if

$$\widehat{F}_{\mathrm{IV}} = F_{\mathrm{IV}}$$

as functions on $\mathbb{Z}_{2^{16}}$.

Because the block size is only 16 bits, this is an exact comparison of two 65536-entry tables.

Theorem 2 (Conditional exactness of the full public attack)**.** *Fix the exact beam IV set used by the outer stage-8 search and the exact fresh-IV sets $\Omega_t(N)$ used by the stagewise search. Assume:*

(a) the outer hybrid bootstrap ranks the true K_8 first;

(b) for each $t \in \{7, 6, 5, 4, 3\}$, on the branch

$$(K_{t+1}, \ldots, K_8) = (K^\star_{t+1}, \ldots, K^\star_8),$$

the true pair $K^\star_t$ belongs to the exact candidate set

$$\mathcal{P}^{(N)}_t(K^\star_{t+1}, \ldots, K^\star_8);$$

(c) for some $\eta \in \Omega_2(N)$ on the true branch, the exact stage-2 scan contains the true row $(K^\star_2, s^\star_3)$, and the stage-1 system

$$W_2(x) = E_1(x + s_1; K_1) + s_2$$

has a unique solution $(K^\star_1, s^\star_1, s^\star_2)$.

Then the implementation outputs the true 256-bit key and accepts it by the exact codebook test. No false full key can be accepted.

Proof. By assumption (a), the recovered outer pair is the true $K^\star_8$. Consider the recursive branch

$$K^\star_7, K^\star_6, \ldots, K^\star_3.$$

At stage t, assumption (b) says that

$$K^\star_t \in \mathcal{P}^{(N)}_t(K^\star_{t+1}, \ldots, K^\star_8).$$

Since the implementation eventually explores every candidate in that set, the true branch is never pruned. Therefore the recursive search reaches stage 2 on the true suffix.

Now fix the stage-2 IV η from assumption (c). On the true branch, the fixed-pair peeling equations (11)–(12) reconstruct the exact public current table

$$C^{(\eta)}_2(x) = R^{(\eta)}_2(x) + s^{(\eta)}_3.$$

Since the exact stage-2 scan includes the true row $(K^\star_2, s^\star_3)$, the reduction (15) holds for the true reduced table

$$W_2(x) = D_2(C^{(\eta)}_2(x) - s^\star_3; K^\star_2).$$

By assumption (c) and Theorem 1, the stage-1 closure returns exactly the unique true triple $(K^\star_1, s^\star_1, s^\star_2)$. Hence the reconstructed full key is the true implementation key. Finally, the acceptance test compares complete 2^{16}-entry tables, so no false full key can be accepted. □

8 Worked random-key recovery

8.1 Target key and successful IVs

As a representative difficult instance, consider the random key

$$K = \texttt{8FCD06ACB99FB6F77135F7ABCD27778AD08489BA2C6DE01CFC17FF0CE0F553C1}.$$

The successful public stagewise run used

- outer beam budget $N_{\text{out}} = 512$;
- beam size $B = 32$;
- search seed 1;
- inward budget $N = 64$ per target stage;
- successful stage-2 IV

$$\texttt{F71831A507C6103EBB51EC0AC07A0C60}.$$

8.2 Recovered values

The public recovery returned

$$\begin{aligned}
K_8 &= (\texttt{E0F5}, \texttt{53C1}), \\
K_7 &= (\texttt{FC17}, \texttt{FF0C}), & s_8 &= \texttt{675F}, \\
K_6 &= (\texttt{2C6D}, \texttt{E01C}), & s_7 &= \texttt{E454}, \\
K_5 &= (\texttt{D084}, \texttt{89BA}), & s_6 &= \texttt{328C}, \\
K_4 &= (\texttt{CD27}, \texttt{778A}), & s_5 &= \texttt{8D1B}, \\
K_3 &= (\texttt{7135}, \texttt{F7AB}), & s_4 &= \texttt{A2CF}, \\
K_2 &= (\texttt{B99F}, \texttt{B6F7}), & s_3 &= \texttt{6DEA}, \\
K_1 &= (\texttt{8FCD}, \texttt{06AC}), & s_2 &= \texttt{7F43}, & s_1 &= \texttt{27B3}.
\end{aligned}$$

The decisive public log line is

```
full-key candidate verified against exact one-word oracle table
```

from `stagewise_attack_hard1_default64.log`.

9 Monte Carlo estimate for full-program random-key success

9.1 Experiment design

This Monte Carlo evaluates the entire implemented attack, not a conditional inward probe. We sampled

$$m = 100$$

independent pseudorandom 256-bit keys by

$$K_j := \text{SHA256}(\texttt{``stagewise-mc-key-}j\texttt{''}), \qquad j = 0, \ldots, 31.$$

For each sampled key we ran the full public program

`separ_public_stagewise_attack.exe`

with

$$N_{\text{out}} = 512, \qquad B = 32, \qquad \texttt{search-seed} = 1, \qquad \texttt{workers} = 16,$$

and inward budgets

$$N \in \{64, 128, 256\}.$$

Because of Proposition 1, success is monotone in N. Accordingly, once a key succeeded at $N = 64$, the larger-budget rows were recorded as monotone skips. The raw data are logged in

`monte_carlo_stagewise_100.log`.

9.2 Observed success rates

For each tested budget, the attack succeeded on all 100 sampled keys. Let

$$P_N := \Pr[\text{full public attack succeeds on a random key at inward budget } N]$$

under the fixed outer parameters

$$(N_{\text{out}}, B, \texttt{search-seed}) = (512, 32, 1).$$

For 32 successes in 32 trials, the exact two-sided Clopper–Pearson 95% lower confidence limit is

$$\underline{P}_{95}(100/100) = 0.963783307\ldots$$

hence

$$P_N \geq 0.9638$$

at confidence level 95% for every tested budget $N \in \{64, 128, 256\}$. Equivalently, the observed 100% success rate carries a one-sided 95% statistical error bar of

$$1 - 0.963783307 \approx 0.036216693,$$

that is, about 3.62 percentage points.

The measured success table is:

Budget N per stage	Total IV contexts	Chosen encryptions	Successes	95% lower bound
64	$512 + 32 + 6 \cdot 64 = 928$	60,817,408	100/100	0.9638
128	$512 + 32 + 6 \cdot 128 = 1312$	85,983,232	100/100	0.9638
256	$512 + 32 + 6 \cdot 256 = 2080$	136,314,880	100/100	0.9638

Because the budget-64 run already succeeded on all 100 keys, the larger-budget rows do not add new observed failures; they certify that success remained perfect when the inward budget was increased.

9.3 Runtime on the Monte Carlo sample

For the 32 full successes at budget $N = 64$, the measured wall-clock times on the user's machine were:

Statistic	Seconds
mean	105.041
median	104.691
min	60.717
max	178.191

Thus, on the tested machine, the default full attack is a seconds-to-minutes computation rather than a brute-force search.

10 Complexity and bound

10.1 Oracle-query complexity

One full one-word codebook requires exactly

$$2^{16}$$

chosen encryptions. The implementation does not require chosen-ciphertext queries, so

$$Q_{\mathrm{dec}} = 0.$$

For outer budget N_{out}, beam size B, and per-stage inward budget N, the successful direct path of the stagewise attack builds:

1. N_{out} outer-search codebooks;
2. B exact beam codebooks for the hybrid K_8 bootstrap;
3. N fresh-IV codebooks for each target stage $t = 7, 6, 5, 4, 3, 2$.

Hence the exact chosen-encryption count on a no-backtracking successful path is

$$Q_{\mathrm{enc}}^{\mathrm{direct}} = \big(N_{\mathrm{out}} + B + 6N\big)2^{16}. \tag{18}$$

With the default parameters

$$N_{\mathrm{out}} = 512, \qquad B = 32, \qquad N = 64,$$

this becomes

$$Q_{\mathrm{enc}}^{\mathrm{direct}} = (512 + 32 + 384)2^{16} = 928 \cdot 65536 = 60{,}817{,}408.$$

10.2 Exact branch-aware bound

The implementation is recursive and may, in principle, backtrack over exact candidate-pair sets before reaching the accepting branch. To state the exact implemented bound, let

$$M_t(N)$$

be the maximum, over all keys and all previously fixed suffixes $(K_{t+1}, \ldots, K_8)$ reachable by the program, of the cardinality of the exact stagewise candidate set

$$\mathcal{P}_t^{(N)}(K_{t+1}, \ldots, K_8).$$

Define recursively the maximum number of fresh-IV codebooks visited below stage t by

$$T_2(N) := N, \tag{19}$$

$$T_t(N) := N + M_t(N)\, T_{t-1}(N), \qquad t = 3, 4, 5, 6, 7. \tag{20}$$

Then the exact program-level chosen-encryption bound is

$$Q_{\mathrm{enc}} = \big(N_{\mathrm{out}} + B + T_7(N)\big)2^{16}, \qquad Q_{\mathrm{dec}} = 0. \tag{21}$$

Equation (21) is the correct exact bound for the implemented recursive search tree. Equation (18) is the realized cost when the true branch is reached without backtracking, which is what occurred on all 32 Monte Carlo keys at budget $N = 64$.

10.3 Exact local work

Once a codebook is built, all remaining work is local and exact. For each stage-t fresh IV, the local work consists of:

1. exact fixed-pair state recovery for each already-solved stage above t, using (10);
2. one exact stage-t low-byte scan;
3. one exact maximizing projected-cycle search over the 2^{20}-entry representative space;
4. exact order reconstruction by Hungarian assignment lifts;
5. exact candidate lifting and exact verifier minimization;
6. for $t = 2$, the exact stage-1 closure of Section 6 together with full-table verification.

10.4 Overall bound

For fixed budgets (N_{out}, B, N), the exact implemented query bound is (21). On the empirically dominant no-backtracking path, it collapses to the direct-path cost (18). For the default parameters of the current binary, the direct chosen-encryption count is

$$60{,}817{,}408.$$

Empirically, on the user's machine, the 100-key Monte Carlo at the default budget $N = 64$ averaged 105.041 seconds per full key recovery, with median 104.691 seconds and worst case 178.191 seconds.

11 Implementation and runtime

The attack is implemented in the multithreaded C program

`separ_public_stagewise_attack.exe`.

The public end-to-end runner is

```
separ_public_stagewise_attack.exe --search-trials N --search-beam B
    --search-seed S --inward-trials M --workers W --oracle-key HEX64
```

with optional `--verbose` output. In the default mode the program shows only the current phase, progress, best IV, current IV, and best recovered key/state values in a compact in-place display.

Operationally, the public path:

- recovers K_8 only by the public hybrid score over the beam;
- recovers stages 7 through 2 only by the exact stagewise recursion of Section 5;
- recovers stage 1 only by the exact closure of Theorem 1;
- accepts a full key only after exact equality of the recovered codebook with the oracle codebook.

The default parameters of the current binary are

$$N_{\text{out}} = 512, \qquad B = 32, \qquad N = 64, \qquad W = 16.$$

These are exactly the parameters used in the full-program Monte Carlo of Section 8.

12 Conclusion

The `main.c` implementation of SEPAR is broken end-to-end in the chosen-IV, chosen-plaintext, chosen-ciphertext, reset model. The decisive structural fact is that every stage is exposed through an exact translated 2^{16}-point permutation, and once a suffix stage key is fixed the corresponding IV-local state word can itself be recovered exactly and peeled away on fresh IVs. This yields the stagewise recursion of Section 5 and removes the dependence on one exceptionally good inward IV.

The attack is therefore not a heuristic key-ranking experiment. Conditional on the sampled IV sets containing the true exact candidates, the recovery is deterministic and exact; false full keys are excluded by complete one-word table equality. On a 100-key Monte Carlo sample of the full program, the default parameter set succeeded on all keys, giving a 95% exact lower confidence bound of 0.9638 on the random-key success probability at the tested budget.